PIONEER JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS 2011

Candidate Name										
Civics Group						**INDEX NUMBER**				

H1 BIOLOGY **8875 / 02**

PAPER 2 **13 September 2011**

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.
This document consists of 2 sections:

Section A
Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B
Answer Question 4.
Write your answer on the lines provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	**/ 10**
2	**/ 13**
3	**/ 10**
4	**/ 7**
Section B	
5 or 6	**/ 20**
TOTAL	**/ 60**

This document consists of **17** printed pages (inclusive of this page)

[Turn Over

Section A: Structured Questions [40 marks]

Question 1

An experiment was carried out to determine what happens to amino acids after they are absorbed by the animal cells. The cells were incubated in a medium containing radioactively labelled amino acid for 5 mins . The radioactive amino acids were then washed off and the cells were incubated in a medium containing only non radioactive amino acids.

Samples of the cells were removed from the cells every 5 mins for 40 mins. For each sample, the level of radioactivity in three different organelle A, B and C were determined.

The results of the experiment are shown in the graph below.

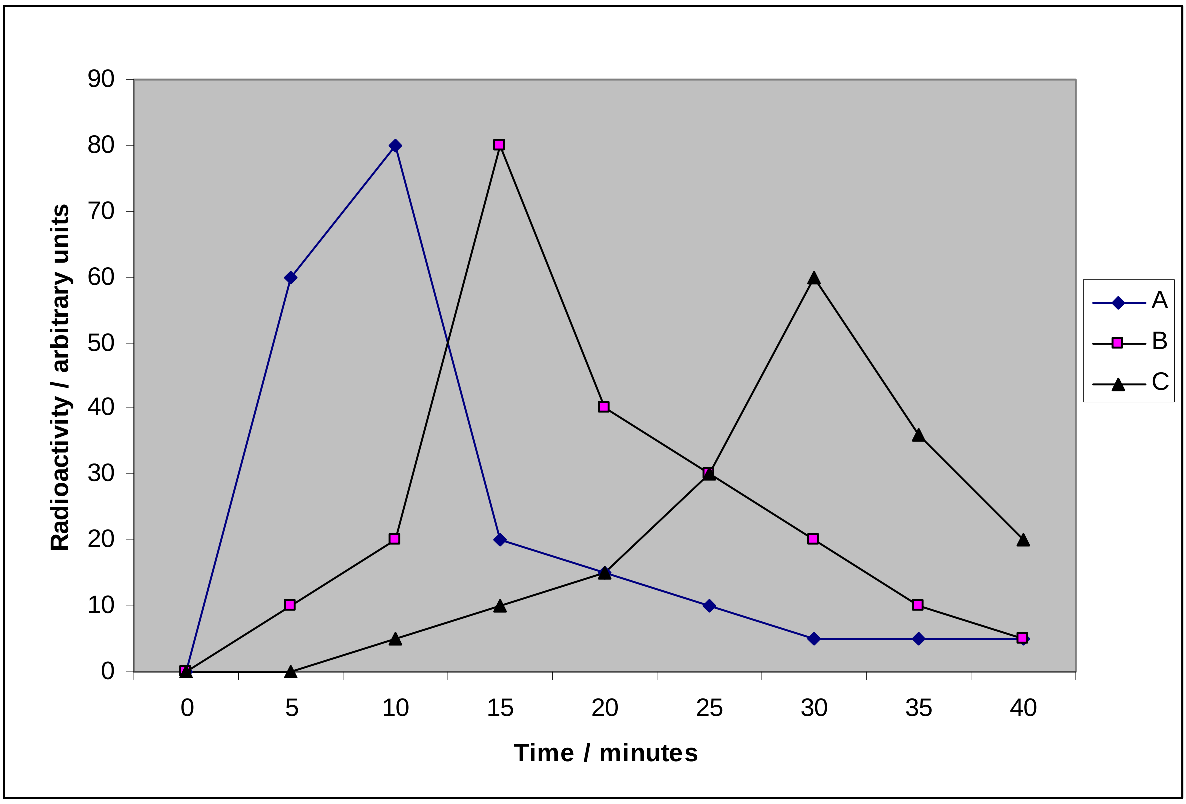

Fig 1.1

(a) Identify the organelle A, B and C by choosing from the list below. Write your answer in the space provided. [3]

Golgi apparatus; nucleus; mitochondria; ribosome; rough endoplasmic recticulum

Organelle A	
Organelle B	
Organelle C	

(b) Explain briefly, with reference to the functions of the organelles, the changes in the levels of radioactivity of the three organelles with time. [4]

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(c) Suggest what will happen to the level of radioactivity after 40 mins and explain your suggestion. [3]

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[Total: 10 marks]

Question 2

Ras gene is an oncogene that codes for a G-protein that relays a signal from a growth factor receptor on the plasma membrane to a cascade of protein kinases.
The cellular response at the end of the pathway is the synthesis of a protein that stimulates the cell cycle.

Mutations in *Ras* gene could lead to the development of cancer.
The organization of the normal **Ras gene** is shown in **Figure 2.1**.

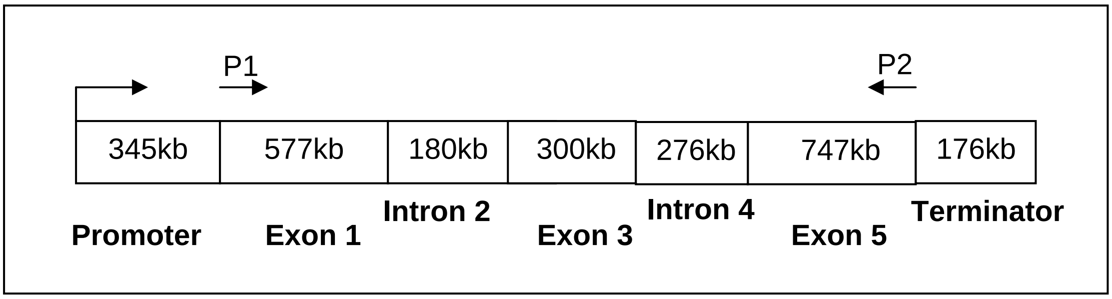

Fig 2.1

Using mRNA from in seven cell types in a cancer patient, reverse transcription was carried out. Subsequently, PCR was performed on the products of reverse transcription using primers P1 and P2, which are specific to the exon regions as shown in Figure 2.1. The PCR products were separated on an agarose gel and the results are shown in the gel image as seen in Figure 2.2.

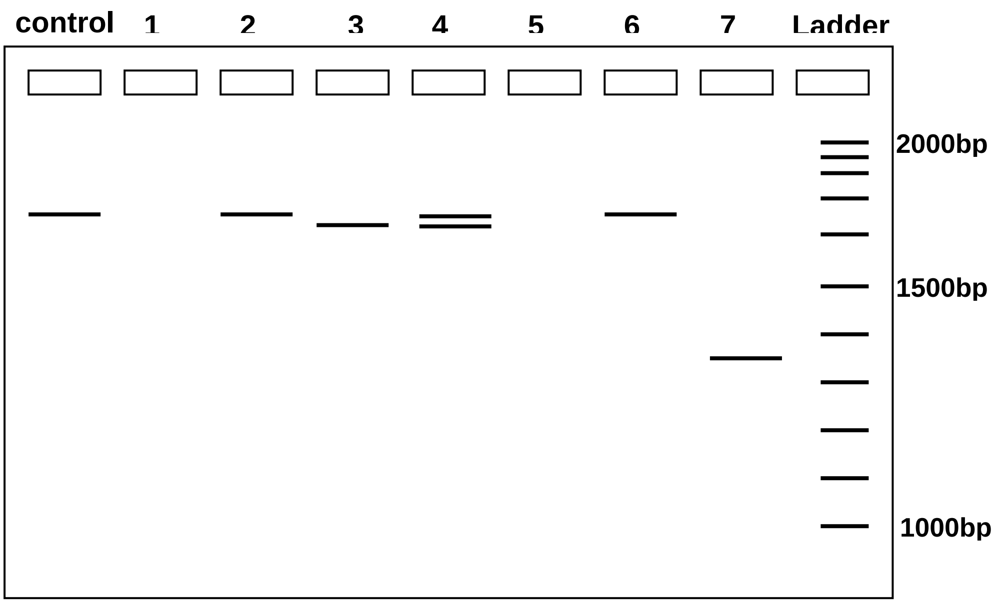

Fig 2.2

Figure 2.3 shows an analysis of Ras protein in the seven cell types in a cancer patient. The control shows the amount of the two substances in a normal cell. A darkly stained band indicates that there is high amount of the respective substance in that particular cell type.

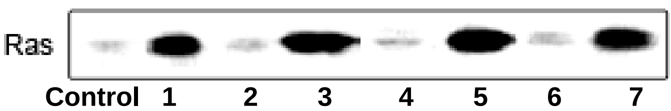

Fig 2.3

(a) Explain how the DNA fragments can be separated by electrophoresis. [2]

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(b) With reference to the gel analysis seen in Figure 2.2 and 2.3, explain how cell type 1 and cell type 5 could become cancerous. [3]

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(c) Cell type 4 was confirmed as a cancerous cell line. With reference to the gel analysis seen in Figure 2.2 and 2.3, suggest an explanation on how cell type 4 could become cancerous. [3]

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(d) Figure 2.4 shows the steps involved in the polymerase chain reaction used to obtain the results as seen in Figure 2.2.

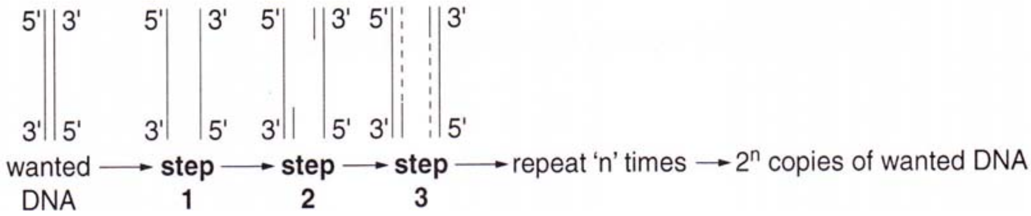

Figure 2.4

With reference to Fig 2.4, describe each of the three steps of one cycle of the PCR.

Step 1……………………………………………………………………………………………

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Step 2……………………………………………………………………………………………

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Step 3……………………………………………………………………………………………

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[Total: 13 marks]

Question 3

Figure 3.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.

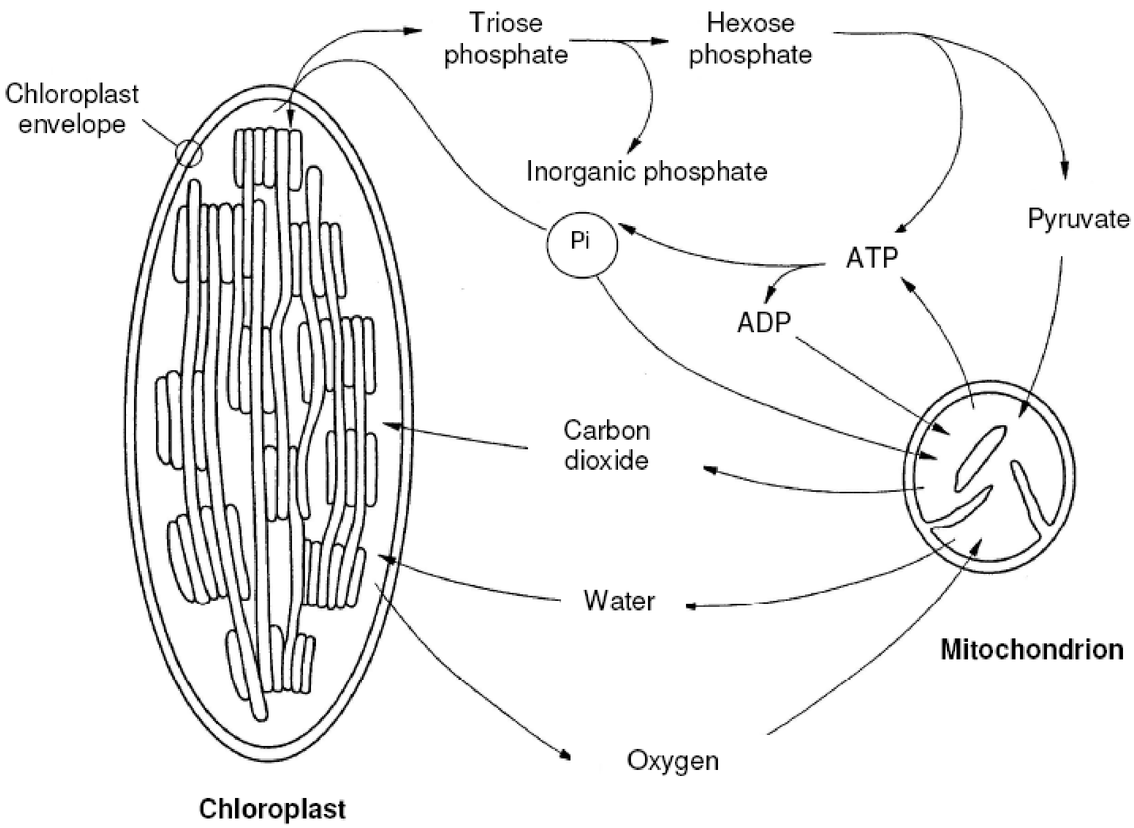

Fig. 3.1

A leafy shoot is sealed inside an air-tight transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Figure 3.1**, outline how the following will link the metabolic pathways between the chloroplast and mitochondrion of this cell:

triose phosphate and hexose phosphate [3],

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(b) With reference to **Figure 3.1**,

(i) explain how it is possible for the oxygen concentration to remain constant. [2]

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(ii) explain why there is no build up in the concentration of phosphate ions inside the mitochondria as a result of the inward passage of phosphate ions. [2]

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An aqueous solution contains small membranous vesicles which may be derived either from a thylakoid or a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell. **Figure 3.2** below shows the various experimental systems that are prepared in an attempt to synthesize ATP.

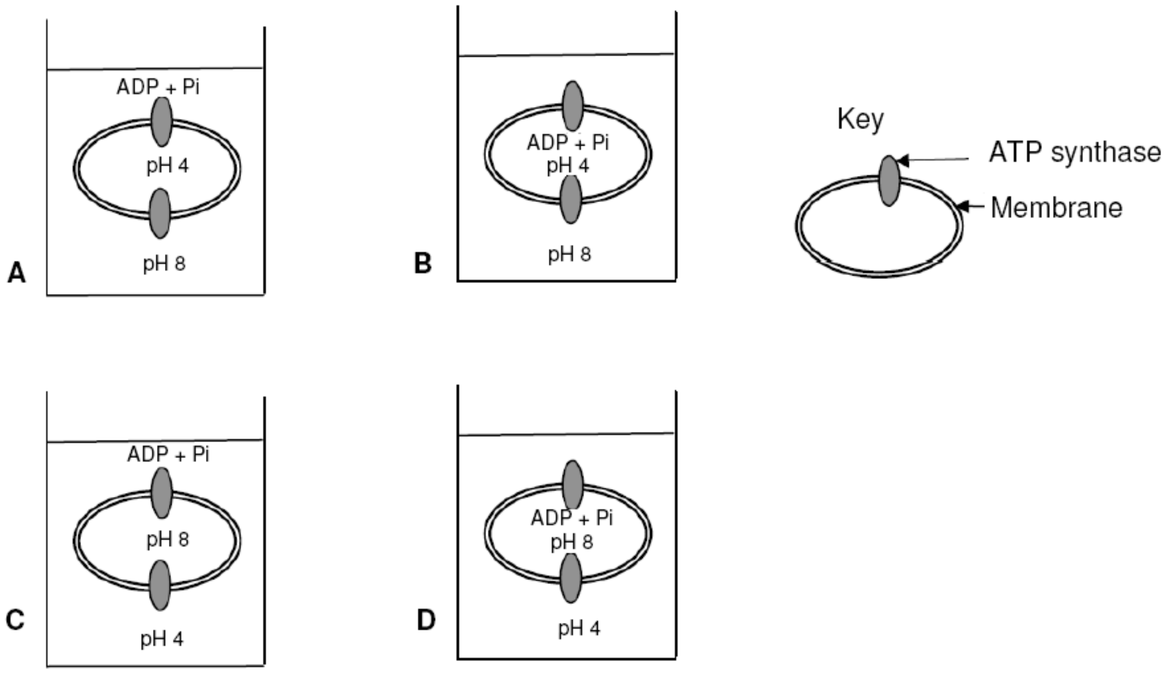

Figure 3.2

(c) Which experimental system (**A** to **D**) shown in **Figure 3.2** would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer. [3]

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[Total: 10 marks]

Question 4

Cytochrome c is found in mitochondria of every aerobic eukaryote – animal, plant and fungi. The amino acid sequences of cytochrome c from many of these organisms have been determined, and comparison made between the molecules shows that they are related.

Human cytochrome c contains 104 amino acids, and 37 of these have been found at equivalent positions in every cytochrome c that has been sequenced. These molecules are homologous.

Table 4.1 shows the first 16 amino acid residues of human cytochrome c, from the N-terminal, with the corresponding sequences from six other organisms aligned beneath.

A ditty (“) indicates that the amino acid is the same one found at that position in the human molecule. All the vertebrate cytochromes (the first four) start with glycine (Gly). The *Drosophila*, wheat, and yeast cytochomes have several amino acids that precede the sequence shown here (indicated by <<<).

Cytochrome c	Amino acid																
		1			4				8				12				16
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln
Pig		“	“	“	“	“	“	“	“	“	“	Val	Gln	“	“	Ala	“
Chicken		“	“	Ile	“	“	“	“	“	“	“	Val	Gln	“	“	“	“
Turkey		“	“	Ile	“	“	“	“	“	“	“	Val	Gln	“	“	“	“
Dogfish		“	“	“	“	“	“	“	“	Val	“	Val	Gln	“	“	Ala	“
Drosophila	<<<	“	“	“	“	“	“	“	“	Leu	“	Val	Gln	Arg	“	Ala	“
Wheat	<<<	“	Asn	Pro	Asp	Ala	“	Ala	“	“	“	Lys	Thr	“	“	Ala	“
Yeast	<<<	“	Ser	Ala	Lys	“	“	Ala	Thr	Leu	“	Lys	Thr	Arg	“	“	“

Table 4.1

(a) With reference to **Table 4.1**, explain what is meant by “These molecules are homologous” (line 7). [2]

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(b) Explain why the analysis of amino acid sequence can be used to establish evolutionary relationship. [3]

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(c) State one limitation using difference in amino acid sequences of only one protein (cytochrome c) to establish phylogenetic relationship among the organisms. [1]

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Cytochrome c is an ancient molecule which has coding and non-coding regions. It has evolved very slowly. Even after more than 2 billion years, one-third of its amino acids are unchanged. The conserved sequences are aa great help in working out the evolutionary relationships between distantly-related organisms like fish and humans. However, for chicken and turkeys, their cytochrome c molecules are identical and evolutionary relationships cannot be determined.

(d) Explain why different species of birds such as chicken and turkey can have the same amino acid sequence in their cytochrome c. [1]

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[Total: 7 marks]

Section B: Essay [20 marks]
Write your answer on the lines provided.

<u>Question 5</u>

(a) Describe the structure of an amino acid and how a peptide bond is formed with another amino acid. [6]
(b) Explain what is meant by primary secondary, tertiary and quaternary structure of a named protein. [8]
(c) Outline how the structure of a named globular protein is related to its specific function. [6]

OR

<u>Question 6</u>

(a) Describe the main features of homologous chromosomes. [6]
(b) Explain how mitosis produces genetically identical cells. [6]
(c) Describe the effect of inhibitors on the rate of enzyme activity. [8]

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